

KACED YUCEF

Computer Security Engineer | Web Development & Cybersecurity | AI & Machine Learning

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PROFESSIONAL SUMMARY

Master's student in Information Systems Security with expertise in cybersecurity, artificial intelligence, secure web development, and geospatial information systems (GIS). Proven ability to design and deploy real-world solutions, including an operational wildfire monitoring platform for Algeria's National Meteorological Office and an award-winning AI system built on data from over 395,000 telecom customers. Recognized through national competitions, including 1st Place at AI EXPO 2026 and 3rd Place at the ITC Challenge. Active Capture The Flag (CTF) competitor with hands-on experience in web security, OSINT, vulnerability assessment, and secure software engineering. Passionate about leveraging technology to solve societal challenges and seeking international research, innovation, and leadership opportunities.

AWARDS & ACHIEVEMENTS

1st Place, AI EXPO 2026 — Université Blida 1 & Djezzy | Ranked 1st out of all Business / General AI Track teams nationally

3rd Place, ITC Challenge Competition 2026 | SahaDz smart healthcare platform for Algeria, selected among national finalists

EDUCATION

Master 1 — Security of Information Systems (SSI)

2025 – 2026 (in progress)

Université Saad Dahleb Blida 1, Algeria

Licence — Information Systems & Quality (SIQ)

2022 – 2025

Université Saad Dahleb Blida 1, Algeria

TECHNICAL SKILLS

Security: Nmap, Burp Suite, Metasploit, Wireshark, OSINT, JWT hardening, threat modeling, vulnerability assessment, phishing detection, browser extension security

AI & Machine Learning: Python, TensorFlow/Keras, CNN, Logistic Regression, Random Forest, NLP, Computer Vision, Streamlit, Pandas

Backend & APIs: Django REST Framework, Java EE, Python, C++, JavaScript, JWT authentication

Frontend: React, JavaScript, Leaflet.js

GIS & Web Mapping: GeoDjango, Leaflet, GeoServer, PostGIS, NetCDF processing, AROME/ALADIN/WRF forecast models

Blockchain: Blockchain architecture design, distributed security, Java EE integration

Databases: PostgreSQL, PostGIS

Systems & Networks: Linux (proficient), Bash scripting, network administration

CERTIFICATIONS

CCNA1

CISCO

PROFESSIONAL EXPERIENCE

Office National de la Météorologie (ONM) — Internship

Dar el Beïda, Algeria | 3 months, 2025

GIS Web Mapping Interface for Wildfire Surveillance & Alerting

- Designed and built a **secured GIS web platform** deployed for ONM meteorologists to visualize and broadcast wildfire risk alerts across Algeria's 48 wilayas
- Integrated meteorological data from **3 forecast models** (AROME, ALADIN, WRF) and 3 fire risk indices (FWI, KBDI, HDW) via NetCDF file processing pipeline
- Architected full client-server security stack (React, Django, PostgreSQL/PostGIS) with **JWT-based authentication** and role-based access control
- Authored complete **technical documentation** covering system architecture, data flows, and security model for operational handoff

Lifeguard — Public Swimming Facility

Summer 2024

- Managed safety and **emergency response** for up to 200 daily visitors, worked under sustained pressure requiring rapid judgment and calm communication with the public
- Developed high **stress-tolerance**, authority under pressure, and responsibility for others — skills directly applicable to incident response and operations roles

PROJECTS

AI-Driven Customer Churn Prediction — Djezzy Partnership

Apr 2025 – May 2026

Stack: Python, Scikit-Learn, Pandas, Streamlit, ML | Université Blida 1 + Djezzy | github.com/Youcef970/CHURN_project

- Built an end-to-end ML pipeline on real-world data from **395,223 Djezzy subscribers** and ~400 behavioral variables to predict customer churn
- Trained and benchmarked Logistic Regression, Decision Tree, and Random Forest models; best model achieved **97% AUC-ROC**, 92.8% Accuracy, and 90.5% Recall
- Addressed severe class imbalance via random undersampling (50/50 balance) and identified the **20 most influential churn indicators** through feature engineering
- Deployed an interactive **Streamlit application** enabling batch prediction, customer profiling, and business-oriented retention strategy analysis
- Awarded **1st Place, Business – General AI Track, AI EXPO 2026**

PhishGuard — AI-Powered Zero-Hour Phishing Defense

February 2026

Stack: Python, FastAPI, Chrome Ext (MV3), Selenium, NLP, CV | CICNIX Cybersecurity Competition | Team: Newbies

- Designed and built a real-time browser extension detecting and blocking **zero-hour phishing attacks** before blacklist databases are updated
- Implemented a **Hybrid AI Engine** combining URL forensics, NLP context analysis, and computer vision-based brand identity verification using Perceptual Hashing (pHash)
- Achieved phishing detection in **under 1.5 seconds** including protection against visual-clone attacks and IDN homograph domains
- Integrated **Chrome Extension** (Manifest V3) with FastAPI backend; automated testing pipeline via Selenium
- Competition track included **Web Exploitation, OSINT**, and Cryptography CTF challenges

SahaDz — Smart Health Platform for Algeria

March 2026

Stack: System Design, Product Management, Business Analysis, Healthcare Tech | ITC Challenge Competition

- Co-developed a **digital healthcare ecosystem** connecting patients, doctors, pharmacies, hospitals, and NGOs to improve medical accessibility across Algeria
- Designed **full platform architecture** covering Doctor Finder (GPS), online booking, personal health records, emergency assistance, and social support services
- Contributed to **monetization strategy**: subscription tiers, freemium model, and institutional partnership framework
- Awarded **3rd Place, ITC Challenge Competition 2026**

GIS Wildfire Alert Interface — Final Year Project (PFE)

2024 – 2025

- End-to-end GIS platform for real-time wildfire risk monitoring across Algeria; **deployed operationally** at ONM during internship
- Secured via **JWT auth**, HTTPS, role-based access, and API input validation; integrated satellite data layers and meteorological risk indices

SECURITY COURSEWORK & TECHNICAL REPORTS

Initiation to Security — Université Blida 1

2025 – 2026

- Conducted and documented **network reconnaissance** using Nmap, Maltego, and OSINT frameworks; formal reports covering attack surface mapping and information gathering methodology
- Produced structured lab reports aligned with **PTES** and **OWASP** testing frameworks

Security of Software Applications — Université Blida 1

2025 – 2026

- Authored vulnerability reports covering **OWASP Top 10**: SQL injection, XSS, broken authentication, insecure deserialization, and privilege escalation
- Applied structured testing methodology from reconnaissance through exploitation to remediation documentation, reports **available on request** (sanitized)

CTF & COMPETITIONS

- **picoCTF** — Active competitor, challenges spanning binary exploitation, cryptography, web security, and forensics
- **Root-Me** — challenges solved across network, web application, and systems categories
- **CICNIX** Cybersecurity Competition — Competed with team Newbies; built PhishGuard under competition conditions alongside CTF challenges in Web Exploitation, OSINT, and Cryptography
- AI EXPO 2026 — **Led team to 1st place** nationally in the Business/General AI Track with the Djezzy churn prediction system
- ITC Challenge 2026 — **Co-led** development and product strategy for SahaDz, awarded 3rd place among national finalists
- FormaXchange Competition — **Led development team** through ideation to prototype under competition time constraints
- INGEHACK 5.0 — Participated in the national cybersecurity Capture The Flag (CTF) competition (April 23–25, 2026) as a member of team Clueless. Focused primarily on OSINT and Web Exploitation challenges under strict “No AI / No LLMs” rules.

LEADERSHIP, VOLUNTEERING & ENGAGEMENT

- **ITC Club Member** — Active member of the ITC Club, contributed to logistics and coordination for large-scale events as both team member and team leader, managing people, timelines, and on-site operations
- **Scientific & Technical Club** — Co-organized scientific conferences and technical workshops, developed project management and cross-functional communication skills
- **Volunteer Work** — Dedicated volunteer during Summer 2020, contributing to the support and well-being of orphaned children through organized activities and sponsorship initiatives.
- **Client-facing** professional roles — Held positions alongside studies requiring direct client communication, team coordination, and service delivery under pressure, building professional reliability and stakeholder management skills

ONLINE PRESENCE & PORTFOLIO

LANGUAGES

Arabic — Native

French — B1 (TCF)

English — B2